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ABBREVIATIONS

CB	Neutron Scattering
CNT	Fracture Wear
RNC	Fatigue Testing

NOTATIONS

F	Force (N)
δ	Displacement (m)

CHAPTER 1

INTRODUCTION

Autonomous vehicles, whether powered by conventional or electric systems, hold immense potential for transforming various aspects of transportation. They offer advantages such as enhanced safety, optimized travel times, and improved energy efficiency, among others. Consequently, in recent years, there has been a notable surge in research and development efforts by universities and automotive companies to realize fully autonomous vehicles. The development of such vehicles typically encompasses different layers or domains, including state estimation and localization, visual perception, and motion planning. Each of these domains presents its own challenges, often involving complex and intricate tasks.

The focus of this manuscript is on motion planning (MP) for autonomous vehicles. Motion planning involves generating a set of instructions that take into account various factors such as the state of the ego-vehicle the surrounding environment (including dynamic and static obstacles, drivable spaces), traffic rules, and allowed maneuvers. The goal of motion planning is to generate a collision-free trajectory and a velocity profile that the autonomous vehicle should follow.

The process of motion planning can be divided into four hierarchical tasks, each assigned specific responsibilities(LaValle, 1999):

1. Mission Planner or Global Path Planning: This task involves determining the high-level mission or goal of the autonomous vehicle. It defines the overall path or route that the vehicle needs to follow to reach the desired destination, considering factors such as road networks, waypoints, and desired objectives.

2. Behavioral Planner: The behavioural planner considers the current driving context and determines the appropriate behaviour or driving strategy for the vehicle. It considers factors such as traffic rules, nearby vehicles, and the overall driving scenario to decide how the vehicle should navigate through the environment. This task helps make high-level decisions such as lane changes, merging, overtaking, or following specific driving patterns.

3. Local Path Planning: Local path planning focuses on generating a collision-free trajectory for the vehicle in the immediate vicinity. It considers the current position, orientation, and dynamics of the ego-vehicle, as well as the presence of obstacles, to compute a feasible path within the local environment. Various techniques, such as sampling-based algorithms or optimization methods, can be employed to generate the optimal path.

4. Local Feedback Control: The local feedback control task is responsible for executing and tracking the planned trajectory in real time. It involves adjusting the vehicle's steering, acceleration, and braking inputs based on sensor feedback and the difference between the planned and actual vehicle states. The control algorithms used in this task aim to maintain accurate tracking of the desired trajectory while accounting for uncertainties and disturbances.

By dividing the motion planning process into these hierarchical tasks, the autonomous vehicle can efficiently navigate its surroundings, ensuring safety and adherence to traffic rules while following the desired trajectory. Each task plays a crucial role in the overall motion planning system, contributing to the successful execution of autonomous driving tasks.

In this research, the primary focus is on behaviour planning and local path planning, as mission planning for a car can often be delegated to tools like Google Maps(Yinglong He, 2019), which can generate a route based on the starting and destination points. While developing the model for the local planner, the local feedback control loop involving actuator dynamics is considered, but it is not the main focus of the research. In the research, actuators are assumed to be first-order black box systems, simplifying their dynamics for the purpose of the study. The emphasis lies in developing efficient algorithms and methods for behaviour and local path planning, critical components of autonomous vehicle motion planning.

The research approach adopted in this study focuses on developing longitudinal and lateral control of the car as separate components and then combining them to work together harmoniously. This approach offers several advantages regarding system development, modification, analysis, and comparison. By treating longitudinal and lateral control separately, researchers can focus on the unique challenges and requirements of each control aspect, allowing for a more detailed and targeted development process.

Separating longitudinal and lateral control facilitates a more modular and flexible system architecture. It enables individual control algorithms to be modified or replaced independently, providing greater freedom to optimize each control aspect based on specific requirements (Yinglong He, 2019). Additionally, this modular approach allows for easy integration of different control algorithms and methods, enhancing the system's versatility and adaptability. Analyzing the system behaviour becomes more manageable when the longitudinal and lateral control components are treated separately. This approach enables effective comparison between different control algorithms or approaches. Practical feasibility is another advantage of this approach.

In the study, the longitudinal planning aspect is approached as the development of adaptive cruise control (ACC). This is accomplished using a Model Predictive Controller (MPC). The details and methodologies related to ACC development are discussed in chapter two of the study.

On the other hand, the lateral controller is developed as a separate component and is the focus of chapter three. In chapter four, the results obtained from the implemented longitudinal and lateral controllers are presented and discussed. These results highlight the performance and effectiveness of the developed controllers in various scenarios and driving conditions. Chapter five of the study addresses the achievements and shortcomings. Additionally, it outlines potential directions for future research and improvements in both the longitudinal and lateral control aspects.

CHAPTER 2

Why Model predictive controller or a Sampling-based approach

2.1 Controllers

In the context of adaptive cruise control (ACC), the modelling of the vehicle does not necessarily require a complex representation. A reasonable level of accuracy can often be achieved by considering a simplified point mass vehicle model that considers system delays, time constants, and first-order vehicle dynamics (Taku TAKAHAMA, 2019). ACC is designed to enhance driver comfort and assist in highway driving scenarios and traffic jams, where the vehicle operates within a range of dynamics that do not push the limits of its capabilities, as seen in high-performance racing scenarios. Therefore, the mentioned approximations and simplifications hold well for ACC, allowing the focus to be on developing effective control algorithms and strategies rather than dealing with the complexities of a detailed vehicle model (Yinglong He, 2019). This simplification enables a practical and efficient implementation of ACC, ensuring its integration into real-world driving scenarios while meeting the objectives of driver comfort and safety.

An alternative approach that can be used for modelling the adaptive cruise control (ACC) system is a transfer function-based method (Yinglong He, 2019). In this approach, a transfer function can be developed to represent the relationship between the desired vehicle acceleration and the control input. A commonly used control algorithm, such as a Proportional-Integral-Derivative (PID) controller, can generate acceleration commands based on the error between the desired and actual vehicle speeds.

However, there are drawbacks associated with this approach. One of the main challenges arises from system delays (Umut Karapınar, 2018), which can lead to undesirable control actions. Due to the inherent time delays in the system, there can be a significant delay between the acceleration commands generated by the controller and their actual implementation. This can result in sudden and abrupt accelerations or decelerations, leading to discomfort for the driver.

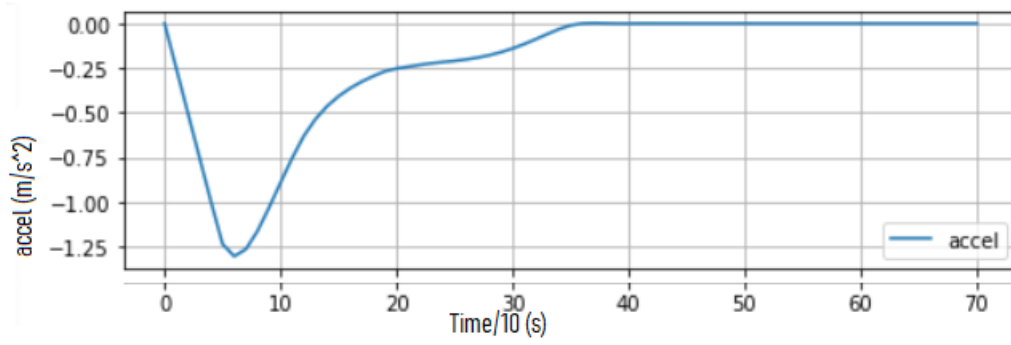


Figure 2.1: The deceleration profile is heavy braking in the beginning, causing discomfort for the drivers.

Additionally, when the ACC system is subjected to constraints such as collision avoidance or following distance requirements, ensuring driver comfort becomes even more challenging. The need to control the actuator limits, such as the maximum acceleration or deceleration rates, can lead to the system underperforming. The aggressive control actions required to meet these constraints can result in jerky and unpredictable vehicle behaviour, negatively impacting driver comfort.

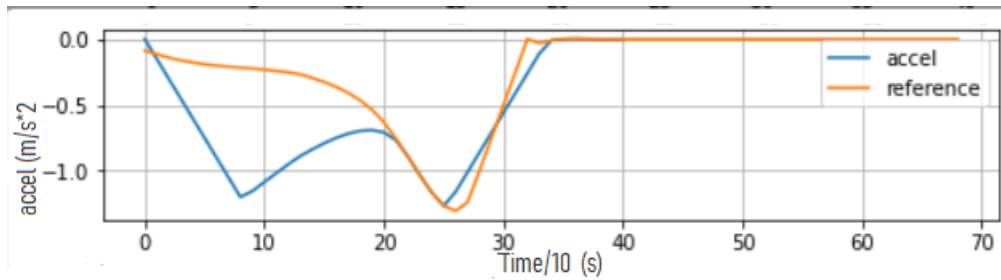


Figure 2.2: The system delays and controller output buildup lead to initial high deceleration, however, as the error from the reference increases again to match the desired profile, the vehicle starts to accelerate, causing discomfort and a sense of unsafety for drivers.

To overcome these drawbacks, advanced control techniques, such as model predictive control (MPC)(Wang, 2009) or adaptive control, can be explored. These approaches consider the system delays and constraints while optimizing control actions over a finite time horizon, resulting in smoother and more comfortable driving behaviour.

An alternative approach to address the mentioned drawbacks and achieve predictive control is through the utilization of sampling-based methods(LaValle, 1999). These approaches offer the capability to ensure adherence to system dynamics, enforce various constraints including environmental and dynamic constraints, and enable the generation of optimal control actions over a prediction horizon.

Next in this chapter, we discuss the formulation of the MPC controller, which aims to achieve predictive control in adaptive cruise control. The mathematical aspects, including the objective function, constraints, and prediction horizon, will be discussed in detail. Later methods based on sampling spaces and trajectory exploration will be examined in the context of lateral motion planning and control.

2.2 MPC - Formulation

In this section, we delve into the use of Model Predictive Control (MPC) to compute spacing-control laws for Transitional Maneuvers (TM) of vehicles equipped with Adaptive Cruise Control (ACC), as proposed by Taku Takahama and Daisuke Akasaka (Taku TAKAHAMA, 2019). The TM problem is formulated as an Optimal Control Problem (OCP), which aims to determine the optimal control inputs to achieve desired spacing between vehicles. The constraints considered in the OCP include the steady-state following distance, collision avoidance, and acceleration limits of the ACC vehicle. By adopting a receding-horizon approach and the spacing-control laws are derived,

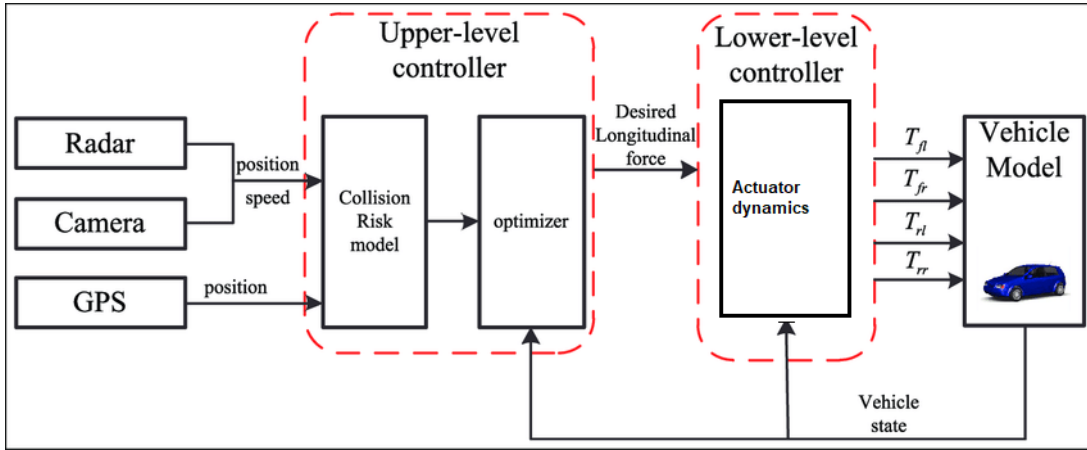


Figure 2.3: Vehicle control architecture includes a low-level controller for immediate control actions (e.g., steering, braking, acceleration) and an upper-level controller for higher-level decision-making (e.g., trajectory planning, obstacle avoidance).

The control system of an Adaptive Cruise Control (ACC) vehicle is commonly structured using a two-level control architecture consisting of an upper-level controller and a lower-level controller (Taku TAKAHAMA, 2019). At the upper level, the controller is responsible for computing the appropriate acceleration commands necessary to execute the Transitional Maneuvers (TM). These acceleration commands are derived based

on factors such as desired spacing-control laws, desired velocity profile, and prevailing environmental conditions. Once computed, the acceleration commands are then transmitted to the lower-level controller. The lower-level controller, known as the tracking controller, receives the acceleration commands from the upper level and converts them into throttle and braking commands.

2.2.1 Problem Definition

The control problem in this paper, denoted as C1-C3, is as follows(Taku TAKAHAMA, 2019):

- C1: Minimize the inter-vehicle distance following error Δd , where $\Delta d = d - d_r$, to converge to zero.
- C2: Minimize the velocity following error Δv , where $\Delta v = v_p - v_h$, to converge to zero.
- C3: Minimize the acceleration of the host vehicle $\dot{v}_h$ to converge to zero.

The objective is to control these errors, ensuring that the inter-vehicle distance, velocity, and acceleration of the host vehicle align with the desired values, resulting in safe and efficient operation of the Adaptive Cruise Control (ACC) system. The actua-

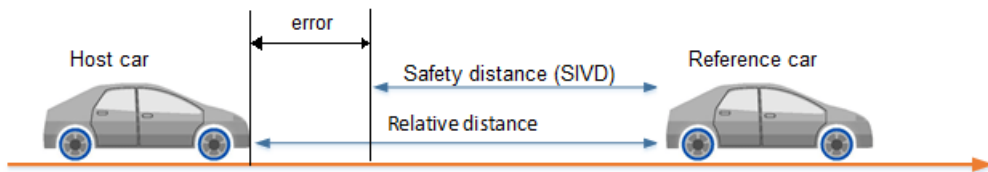


Figure 2.4: Representation of safety distance (SVID), relative distance and tracking error between Host and reference vehicle. The error specifically refers to the distance error between them.

tion dynamics of a vehicle, which includes the engine, transmission, and brake, exhibit nonlinear characteristics. Typically, the relationship between the input and output of the actuation system is described by an ordinary differential equation. The spacing-control laws is designed based on the following first-order ACC model:

$$\tau \frac{d^2x}{dt^2} + \frac{dx}{dt} = u(t), \quad \frac{dx}{dt} \geq 0, \quad u_{\min} \leq u \leq u_{\max}$$

where $x(t)$ represents the inter-vehicle distance, $u(t)$ represents the control input, and $u_{\min}$ and $u_{\max}$ are the minimum and maximum control limits, respectively. The system dynamics are captured by the second derivative of the inter-vehicle distance, $x''(t)$, and the time constant is denoted by τ . The control objective is to determine the optimal control input $u(t)$ that satisfies the system dynamics while ensuring that the inter-vehicle distance is maintained within the desired range.

2.2.2 State Space Formulation

To build a plant model for the ACC system design, two state variables are defined: inter-vehicle distance following error $\Delta d = d - d_r$ and velocity following error $\Delta v = v_p - v_h$. The d_r is determined based on the constant time headway policy given by $d_r = T_{hw} \cdot v_h + d_0$

where T_{hw} is the constant time headway, and d_0 is the stopping distance for a safety margin.

Let us define the state variables of the plant as $x = [x_1 \ x_2 \ x_3]^T \in R^{2+n_f}$ with $x_1 = \Delta d$, $x_2 = \Delta v$, and $x_3 = x_f$.

$$\dot{x} = A(t)x + B(t)u, \quad y = Cx,$$

where $\dot{x}$ represents the derivative of x with respect to time, u is the control input, and y is the output.

$$A_f(t) = -\frac{1}{T_{eng}}, B_f(t) = -\frac{K_{eng}(t)}{T_{eng}}, \text{ and } C_f = 1.$$

Here, T_{eng} represents the time constant of acceleration using the engine, and $K_{eng}(t)$ is the time-dependent steady-state gain. In the system, the dynamics are separated into the acceleration and deceleration sides, and they are modelled as a first-order delay system. The state space model in continuous

$$x \in R^3 \quad \text{and} \quad A(t) = \begin{bmatrix} 0 & 1 & -T_{hw} \\ 0 & 0 & -1 \\ 0 & 0 & A_f(t) \end{bmatrix}; \quad B(t) = \begin{bmatrix} 0 \\ 0 \\ B_f(t) \end{bmatrix}$$

space is In this model, the rank of matrix A is 2, indicating that the system is not fully

controllable. This implies that not all inputs can drive all states to zero or any desired state. Since only the host velocity can be directly controlled, the controllability of the system is not of significant importance. The additional state variable, Δv , is influenced by the host and preceding velocities, v_h and v_p respectively.

The system is linear and hence can be directly linearized (Wang, 2009).

$$\begin{bmatrix} x(t+T) \\ \dot{x}(t+T) \\ \ddot{x}(t+T) \end{bmatrix} = \begin{bmatrix} 1 & T & 0 \\ 0 & 1 & T \\ 0 & 0 & 1-T\tau \end{bmatrix} \begin{bmatrix} x(t) \\ \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ T\tau \end{bmatrix} u(t)$$

where T represents the sampling time of the system.

2.2.3 Feasible Initial Conditions - SIVD

Initial conditions refer to the values of R , $\dot{R}$, and ACC vehicle velocity and acceleration when the target is first encountered. The state variable R of ACC is defined relative to the Specified Inter-Vehicle Distance (SIVD), which the driver determines. A scenario is considered feasible if the ACC vehicle can achieve the target vehicle velocity by decelerating from its initial velocity while avoiding collision with the target vehicle. By solving the equations, we can identify the initial conditions that satisfy the feasibility criteria.

$$\begin{aligned} x(t) &= \begin{bmatrix} x(t) \\ \dot{x}(t) \\ \ddot{x}(t) \end{bmatrix} \\ x(t) &= e^{At}x(0) + \int_0^t e^{A(t-\eta)}Bu(\eta)d\eta \end{aligned}$$

where 0 and t are set as the initial and final times of the maximum deceleration manoeuvre, respectively. The initial conditions can be written as:

$$x(0) = \begin{bmatrix} 0 \\ \dot{x}(0) \\ -v_0 \end{bmatrix}$$

Given e^{At} and $x(0)$ for the maximum deceleration manoeuvre, the solution of the

system of equations can be written as(Wang, 2009):

$$\begin{aligned}x(t) &= (\dot{x}(t) - v_0)t + \left(\frac{1}{4} - \frac{t^2}{2} + \frac{t^2}{2}e^{-2t}\right)u_{min} \\ \dot{x}(t) &= \dot{x}(0) - v_0 + \left(-\frac{1}{2} + t + \frac{1}{2}e^{-2t}\right)u_{min} \\ \ddot{x}(t) &= (1 - e^{-2t})u_{min}\end{aligned}$$

The SIVD is calculated by finding the value of $x(t)$ and t at which the range rate becomes zero. If the initial range is greater than the obtained SIVD, the initial condition is said to be feasible.

2.2.4 Formulation for MPC

The control objectives and other requirements of the control problem are formulated as a Fixed Time Constrained OCP (FTCOCP) as follows(Wang, 2009) (Taku TAKAHAMA, 2019). With an initial error of $x(0)$, a control sequence is determined that minimizes the performance index

$$J = x_N^T S x_N + \sum_{k=0}^{N-1} (x_k^T Q x_k + u_k^T R_{N-1} u_k)$$

$x(0) \equiv$ measured, with the state and control constraints which formulate the collision avoidance, non-negative velocity constraint, and acceleration limits of the ACC vehicle respectively

$y_k = (x_k - \dot{x}k) \leq SIVDv_0 = Sc.$ $u_{min} \leq u_k \leq u_{max}$. The state feedback spacing-control law is computed using the receding-horizon approach. The range, range rate, ACC vehicle velocity, and accelerations are measured at each sampling time t , and the SIVD is updated. Then, the error vector is computed. Using the error vector as the initial condition, the QP is solved for the entire time horizon:

$$\min_U U^T H U + 2G(x(t))U$$

$$\text{Subject to } L_{IN}U \leq M_{IN}(x(t)) \text{ and } L_{EQ}U = M_{EQ}(x(t))$$

to obtain an open-loop control sequence U_{tol} .

CHAPTER 3

Sampling-based Lateral Motion planning

3.1 Methodology

Local path planning has been extensively researched, leading to the development of various efficient algorithms. These algorithms can be categorized into three main groups: Graph Search Algorithms, Random Sampling Algorithms, and Intelligent Bionic Algorithms. Graph Search Algorithms, such as the Dijkstra algorithm, A* algorithm, BFS algorithm, and DFS algorithm, have been widely used in path planning. However, these methods may face limitations in terms of planning efficiency when dealing with complex environments.

Random Sampling Algorithms, including BIT, RABIT, RRT, and Risk-DTRRT, offer improved efficiency and are commonly used in dynamic high-dimensional environments. These algorithms efficiently explore the search space by sampling potential configurations, enabling effective path planning. Intelligent Bionic Algorithms, such as GA, ACO, ABC, and PSO, simulate evolutionary and biomimetic behaviors to find locally optimal paths. Techniques like optimization of genetic algorithm-particle swarm optimization algorithm, combined approaches with artificial potential field and geometric local optimization, and constrained multiobjective PSO with accelerated update methodologies have been proposed to enhance the search for locally optimal paths.

In the context of local planning, the A* algorithm is implemented. This choice is based on the fact that the driving environment primarily consists of highway scenarios, which are relatively less complex compared to other environments. The A* algorithm is well-suited for finding efficient paths in such situations (Thomas M. Howard and Kelly, 2008).

Additionally, we assume a known reference trajectory that we want to track. This trajectory is obtained from sources like Google Maps and is represented by data points along the centerline of the road.

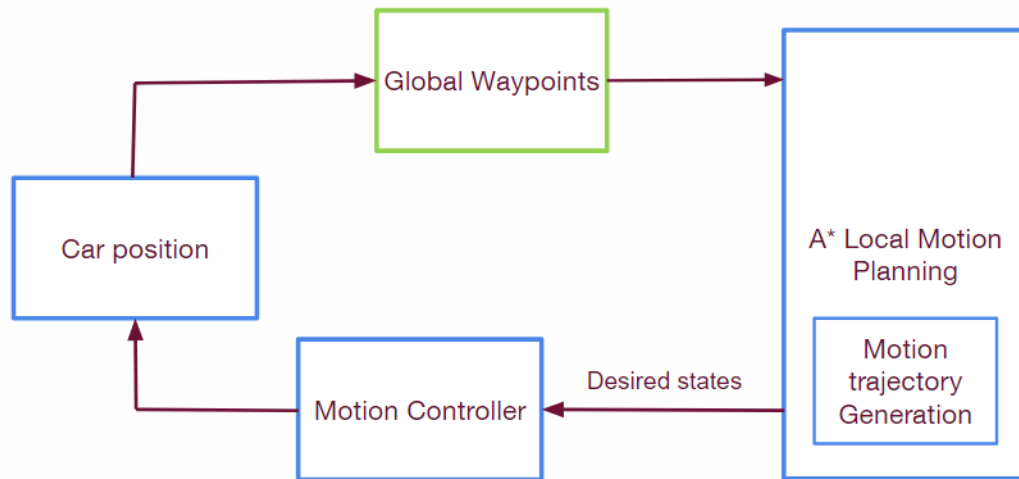


Figure 3.1: The block diagram outlines the methodology for developing a motion planning algorithm

3.1.1 Motion Trajectory Generation

The local motion planner plays a critical role in obstacle avoidance within the immediate surroundings, even when following a global path. To address obstacles along the path, motion trajectory generation techniques are employed. Two main methods for solving this problem are state space sampling and control space sampling.

State space sampling involves generating different states or configurations of the vehicle within the environment. These states are sampled to find a trajectory that avoids obstacles and satisfies the desired objectives. The planner searches for feasible states while considering constraints such as vehicle dynamics, collision avoidance, and path smoothness.

On the other hand, control space sampling focuses on generating control inputs or actions for the vehicle. Instead of sampling different states, this approach explores various control inputs to find a trajectory that avoids obstacles and achieves the desired behavior. The control inputs are adjusted iteratively to satisfy the constraints and improve the trajectory quality.

Control space trajectory generation is a method used to address the local motion planning problem. Unlike the state space approach, which involves optimizing trajectories, control space sampling offers a lightweight alternative. In the state space approach,

defining what constitutes an optimal trajectory can be challenging, and it often requires tuning parameters and computationally intensive calculations.

Control space sampling, on the other hand, simplifies the process by exploring control inputs directly. By sampling different control inputs, the algorithm can quickly generate trajectories that avoid obstacles and meet the desired objectives. This lightweight approach enables larger look-ahead predictions, allowing the planner to consider a longer time horizon when making decisions.

The general framework of the A* algorithm

After a state with the least cost is popped out of the closed fringe.

1. Next state (neighbour) generation occurs.
2. For each neighbour generated the node id is appended
3. Cross-track error and path length is calculated, the node is discarded if cross track error is more than the track width
4. Check for obstacles; node discarded if with a range of obstacle
5. Check for goal state.
6. Costs are calculated
7. If nodes have x, y coordinates, theta and delta within error resolution, then the node is discarded.
8. A neighbouring node is added to the fringe.

3.2 Control Space Sampling

Control space sampling expands the traditional x, y coordinate framework by adding two additional axes (Julian P. Timings¹, 2013): distance along the center line (ds) and track orientation (ϕ). Initially, the A* algorithm was developed solely using x, y coordinates, but certain challenges emerged.

In the x, y coordinate-based implementation, each state includes x, y coordinates, vehicle heading angle, steering angle, parent ID, and cost. To account for continuous

sampling in space, states with small differences are considered the same. Feasible trajectories for the vehicle are generated using the kinematics of a bicycle model, resulting in neighbor states with varying x , y , θ , and δ .

However, this formulation has drawbacks. The algorithm requires the generation of a goal state, which should be a constant look-ahead time away from the current position and ideally on the center line. Curve fitting is necessary to determine the goal state coordinates. Calculating cross-track error involves finding the normal to the curve and its length through the state coordinates. Checking if a state is inside the track boundary also requires computation of the track boundary curve, which needs to be dynamically computed. Additionally, determining orientation error and distance traveled along the center line involves complex mathematical operations on the center line curve.

To address these challenges, an alternative formulation using additional ds and ϕ coordinates is proposed. The track is defined using x, y, ds (distance along the center line), and ϕ (heading angle) at the current point. The state definition now includes the current distance traveled by the vehicle. Key differences from the previous formulation include:

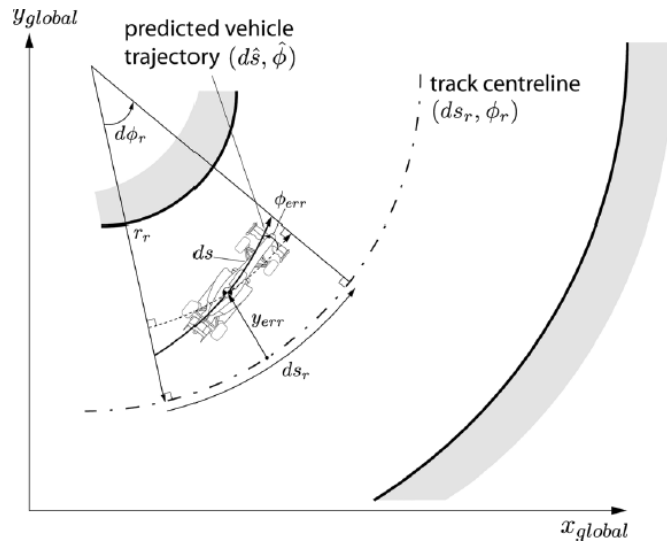


Figure 3.2: ds and ϕ axis, the heading angle error ϕ_{err} , distance travelled in a time step (ds) and the cross track error is represented (Julian P. Timings1, 2013)

1. For each generated trajectory, the length of the trajectory is determined. This length is traversed along the track centre line, and the shortest normal from a few nearest points to the current trajectory provides the coordinates along the center line.
2. The cross-track error and distance error are obtained based on the above operation.

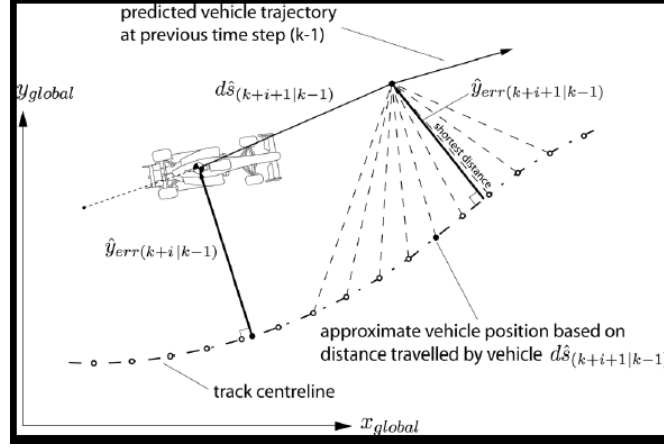


Figure 3.3: Finding nearest points along the centre line(Julian P. Timings1, 2013)

3. Theta error is calculated as the difference between the state's θ and the track's θ .
4. The goal point can be easily defined for a look-ahead time, given the distance traveled is $v \cdot T_{\text{heading}}$.

The inclusion of ds and ϕ coordinates simplifies trajectory generation and reduces the reliance on curve fitting operations. It enables more efficient calculation of cross-track error, distance error, and theta error. Furthermore, the goal point can be readily determined for a given look-ahead time. This enhanced formulation improves the effectiveness and efficiency of the control space sampling approach for local motion planning.

3.2.1 Motion model

The bicycle motion model, as depicted in Figure 3.4, to describe the motion of the vehicle. This model assumes that the vehicle behaves similar to a bicycle, with a single front wheel and two rear wheels. These assumptions are valid for a large velocity range with small computation steps (i.e. slip should not be large for a time step).

The bicycle motion model consists of several equations that govern the vehicle's dynamics. These equations relate the vehicle's state variables, such as position, orientation, and velocity, to its control inputs, namely steering angle and velocity. The specific equations used in the bicycle motion model include:

Kinematic Equations:

$$\dot{x} = v \cdot \cos(\theta)$$

$$\dot{y} = v \cdot \sin(\theta)$$

$$\dot{\theta} = \frac{v}{L} \cdot \tan(\delta)$$

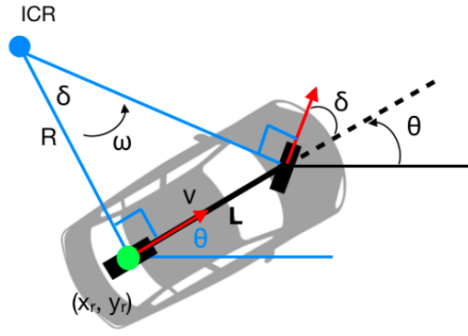


Figure 3.4: Kinematic bicycle model

Rate of change of the x-coordinate (position along the x-axis) is determined by the vehicle's velocity v and the cosine of its orientation angle θ . Rate of change of the y-coordinate (position along the y-axis) is determined by the vehicle's velocity v and the sine of its orientation angle θ . Rate of change of the orientation angle θ is determined by the vehicle's velocity v , the wheelbase length L (distance between the front and rear axles), and the tangent of the steering angle δ . **Control Inputs:**

Steering Angle (δ)

Velocity (v)

The Steering angle of the front wheel with respect to the vehicle's longitudinal axis. It determines the direction in which the vehicle will turn. Velocity (v): The speed of the vehicle. For each planning cycle the current velocity of the vehicle will be obtained from the longitudinal motion planning and will remain constant for one cycle of lateral motion planning.

3.2.2 Formulation

In the control space formulation, the distance D forward is given by the product of the current velocity v and the time horizon T .

To generate the next feasible state, we perturb the steering angle δ from the current state. Each generated state corresponds to the forward kinematics of the bicycle model for one prediction step, or one time step. The state is represented by the variables:

$$\{x, y, \theta, \delta, \text{parent_id}, \text{cost}\}$$

where (x, y) represents the position, θ represents the heading angle, δ represents the steering angle, parent_id represents the parent node id, and cost represents the total cost of the current node.

The goal is to travel the distance D forward, which means that the final state should have a position (x_f, y_f) such that the distance between (x, y) and (x_f, y_f) is equal to D .

The costs associated with each state include the cross-track error y_{err} , the distance travelled ds , and the change in steering input $\Delta\delta$. These costs are taken into account to evaluate the desirability of each state.

The heuristic used in the A^* algorithm is defined as $H = D - ds$, where ds represents the distance along the centerline. The distance along the centerline is calculated by summing the distances between consecutive points on the centerline.

CHAPTER 4

Results

4.0.1 ACC - Test Scenarios

A Comprehensive Set of Test Scenarios for Evaluating Adaptive Cruise Control Algorithm Performance in Simulated Highway Driving Conditions. These scenarios are made to simulate the essential dynamic requirements encountered during highway driving scenarios. The evaluation of the adaptive cruise control algorithm's performance is based on the ability of the vehicle to navigate and respond to the various scenarios presented effectively(Chen, 2022).

Note: All the graphs have SI units along the axis unless specified otherwise.

Scenario 1

The test scenarios encompass situations where the car is initially stationary, maintaining a certain distance from the reference vehicle. Subsequently, the reference vehicle initiates movement. These specific scenarios are designed to evaluate two essential capabilities of the vehicle. Firstly, it assesses the vehicle's ability to gradually approach and achieve the desired position relative to the reference vehicle within traffic conditions. Secondly, it evaluates the vehicle's capacity to accelerate smoothly and maintain a consistent pace following a complete stop.

Scenario 2

The test scenarios presented consist of a dynamic sequence of events. Initially, both car remains stationary while the vehicle in front commences movement with a constant acceleration. Following this, the lead vehicle maintains a consistent velocity for a certain duration before decelerating and eventually coming to a complete stop. These versatile scenarios are designed to simulate a wide range of conditions, enabling a comprehensive evaluation of the stop-and-go controller's full capabilities.

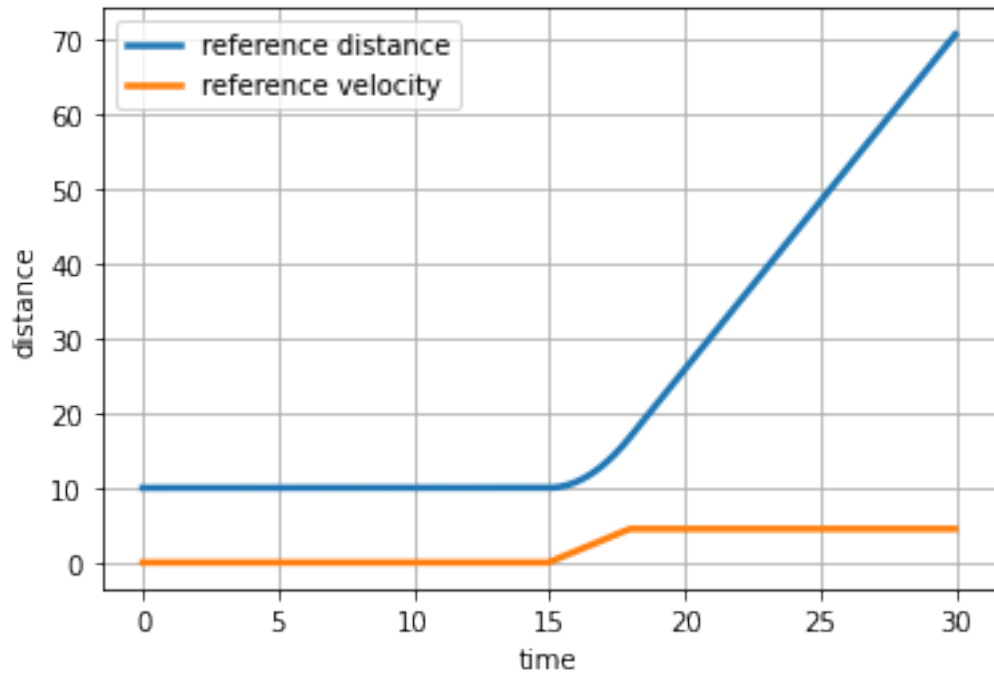


Figure 4.1: The host vehicle is stopped behind the reference vehicle with an initial distance of 10 meters. Then the reference vehicle accelerates

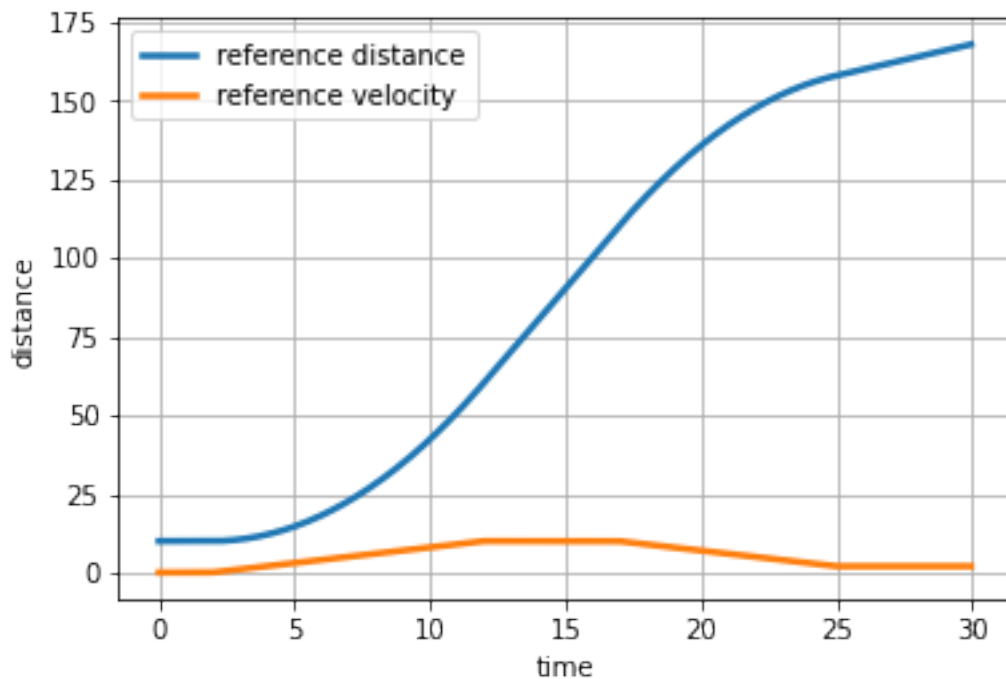


Figure 4.2: Scenario 2: Initial stationary vehicles, followed by acceleration and deceleration of the reference vehicle. The reference vehicle achieves a desired velocity, maintains it for a duration, and eventually comes to a stop.

Scenario 3

In this specific scenario, the motion of the reference vehicle follows the same pattern as described in scenario 2 (initial acceleration, constant velocity, and subsequent decel-

eration). However, what sets this scenario apart is the intentional variation in the state space parameters of the system. The primary dynamics of interest in this scenario are acceleration and deceleration, and the time constants and input gains of the low-level controller play a crucial role in shaping the vehicle's response.

By introducing variations in the modeling parameters, this scenario aims to evaluate the robustness of the Model Predictive Control (MPC) system. Robustness refers to the ability of the MPC algorithm to maintain satisfactory performance and stability despite uncertainties or changes in the underlying system dynamics.

Scenario 4

This scenario is specifically designed to validate the performance of the vehicle in overtaking. In this scenario, the host vehicle is travelling at a constant speed and the reference vehicle is introduced at a look-ahead distance from the host vehicle. The reference vehicle also maintains a constant speed but occasionally experiences small accelerations. The objective of this scenario is to assess the host vehicle's ability to respond to the presence of the reference vehicle, adjust its speed accordingly, and maintain a safe distance while accommodating the reference vehicle's intermittent accelerations. This scenario encompasses two variations: a higher-velocity vehicle overtaking and a lower-velocity vehicle overtaking

4.0.2 ACC - Results

Scenario 1

The observation of the results can be depicted in (4.4). The plot illustrates the successful tracking of the reference vehicle by the host vehicle, as the initial gap is closed, and a safe distance is maintained to avoid tailgating. However, when the reference vehicle unexpectedly accelerates, a delay in the host vehicle's response is observed. This delay can be attributed to the dynamics of the low-level controller and the adjustment of the safety distance.

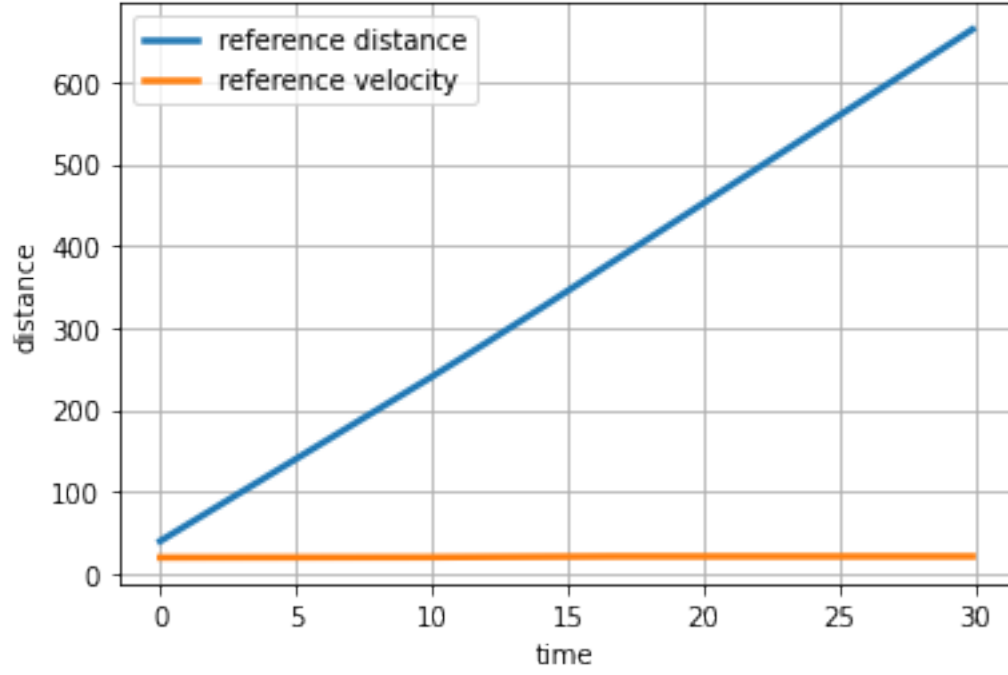


Figure 4.3: Scenario 4: Highway overtaking

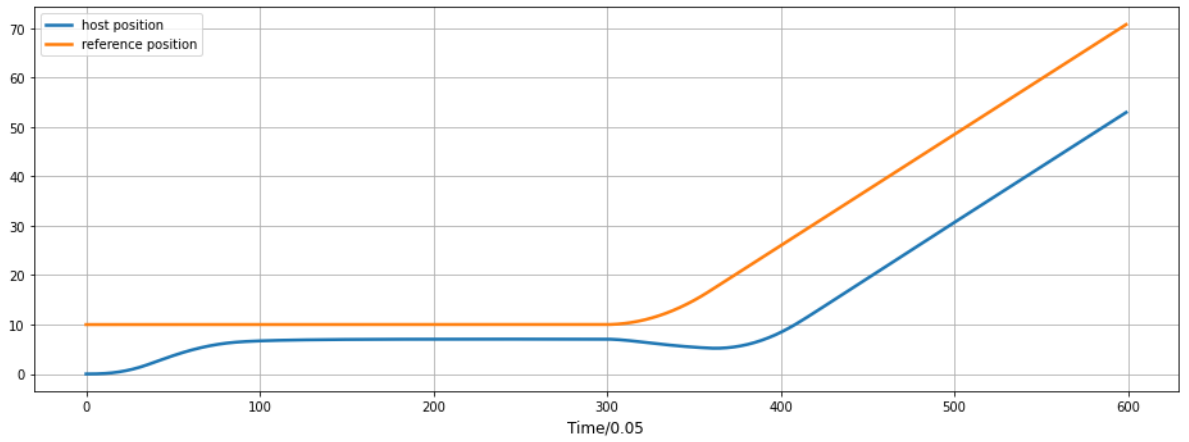


Figure 4.4: Position of reference and host vehicle for scenario1, It can be noticed that the host is able to track the profile reasonably

Scenario 2

The results shown in Figure 4.9 indicate that the controller is capable of closely tracking the reference vehicle in a traffic scenario and maintaining a constant safety distance with the reference vehicle.

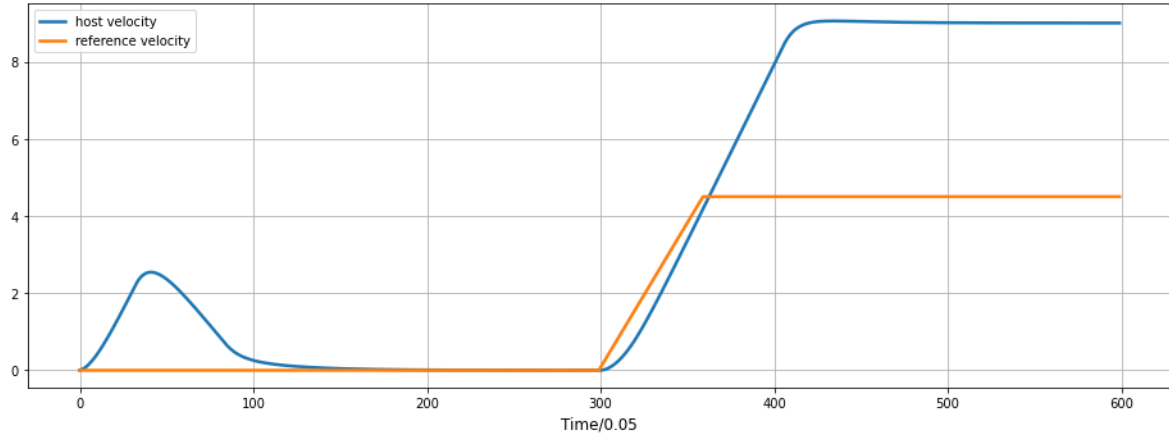


Figure 4.5: The velocity profile of the reference and host vehicle. The vehicle tries and archives zero velocity. In the later portion, the velocity of the host is higher.

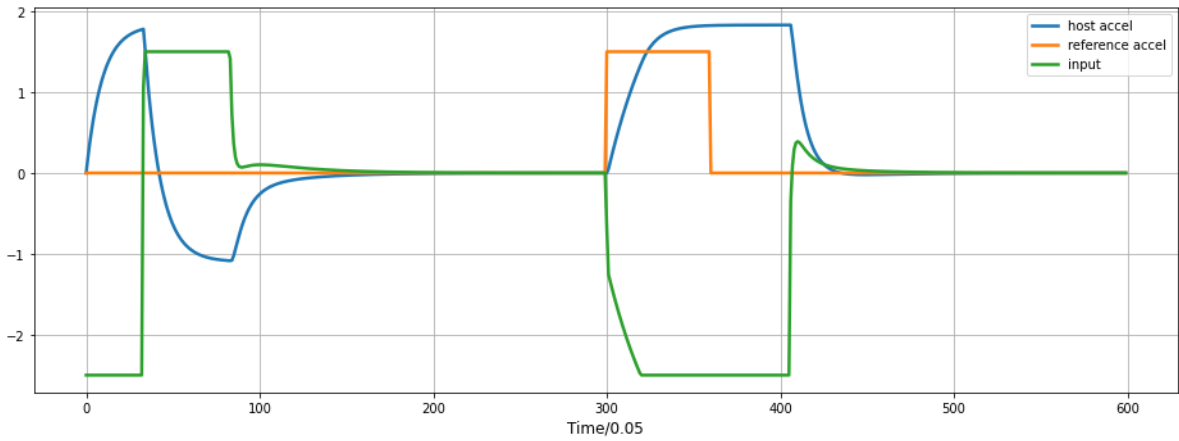


Figure 4.6: The analysis of the control input profile reveals that during the transitions in the scenario, the controller is fully actuated, indicating that it exerts maximum control effort to execute the desired actions. Additionally, the acceleration commands of the host vehicle follow first-order dynamics, aligning with the underlying modelling assumptions.

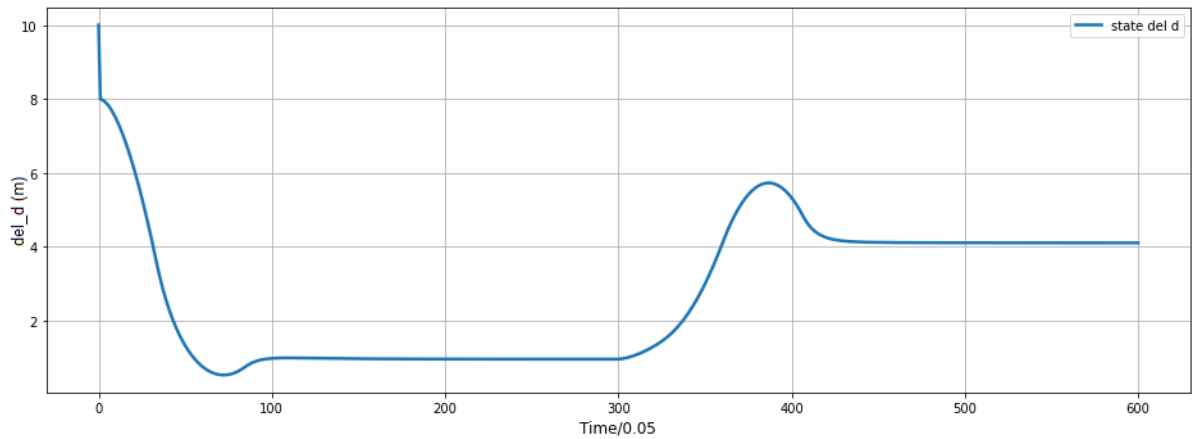


Figure 4.7: Variation of state parameter relative distance is as shown.

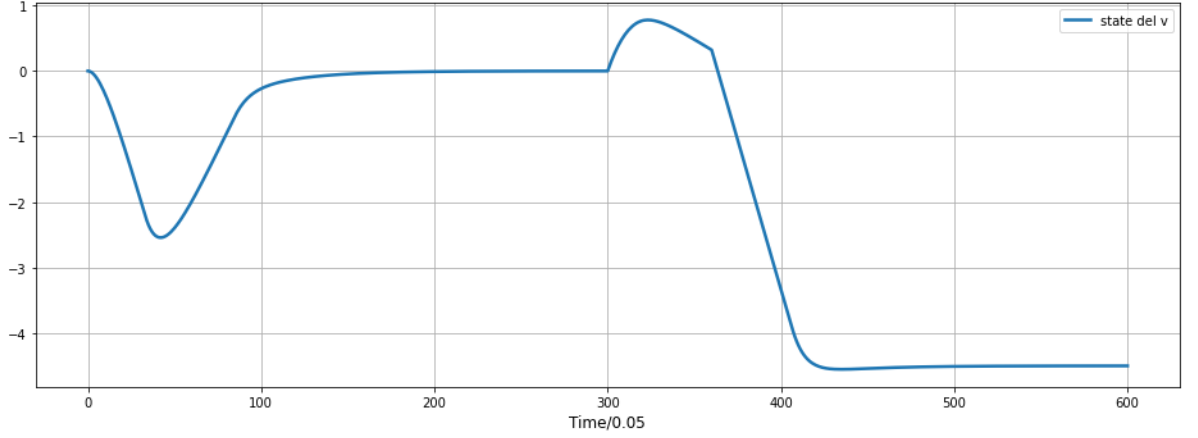


Figure 4.8: Variation of state parameter relative velocity is as shown.

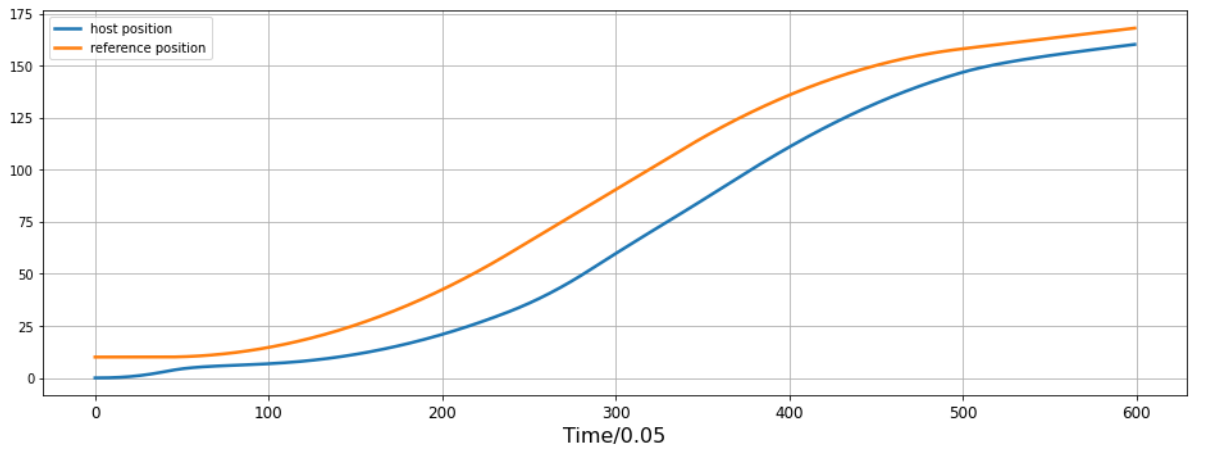


Figure 4.9: The host vehicle is able to closely track the reference vehicle even in continuous acceleration and deceleration scenarios.

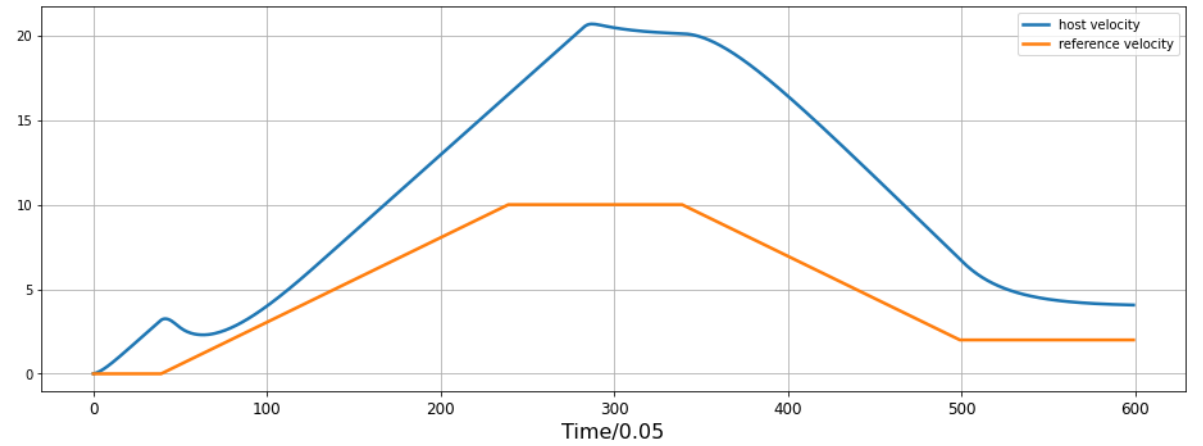


Figure 4.10: The velocity profile doesn't match exactly with the reference profile, but it is to be noticed that the final velocity is converging and the position tracking is achieved.

Scenario 3

The time constant of acceleration dynamics is varied by 20%. The simulation results show that trajectory tracking is achieved without major deviations from the previous

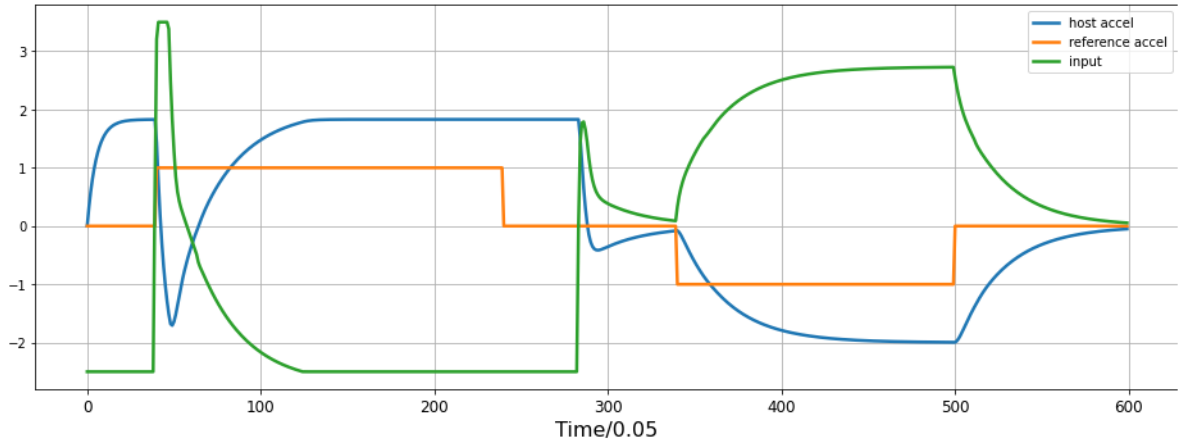


Figure 4.11: The acceleration profile depicts controls actions are fully unitized while there is an abrupt change observed in states, it is reasonable as these are extreme scenarios .

profile (Figure 4.9). This depicts the robustness of the controller towards changes in low-level controller dynamics, which are often encountered in real-life scenarios.

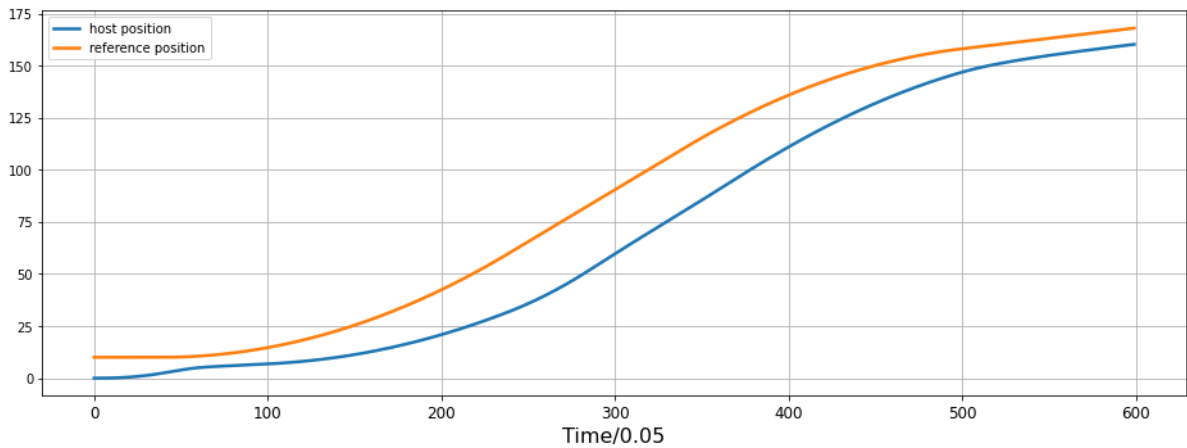


Figure 4.12: The position tracking closely matches with the scenario 2 results, but minor changes exist and can be noticed while examining the controller action profile .

Scenario 4

The results of the higher velocity vehicle overtaking scenario are illustrated in Figure 4.14. It is evident from the plot that the host vehicle responds appropriately when a higher-velocity vehicle is introduced ahead of it, successfully catching up to its speed. The host vehicle initially travels at a velocity of 54 Kmph, while the overtaking vehicle maintains a speed of 72 kmph. But it is worth noting that in all scenarios host vehicle is not able to do so and requires tuning of weights to achieve optimal distance and velocity

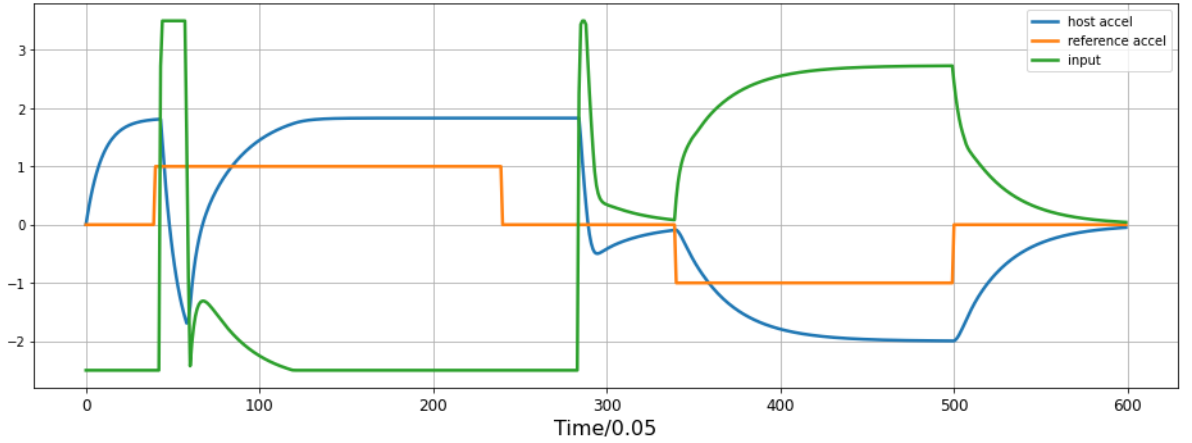


Figure 4.13: There is a noticeable difference between the controller actions on the change of time constant, as there are higher controller actions.

tracking.

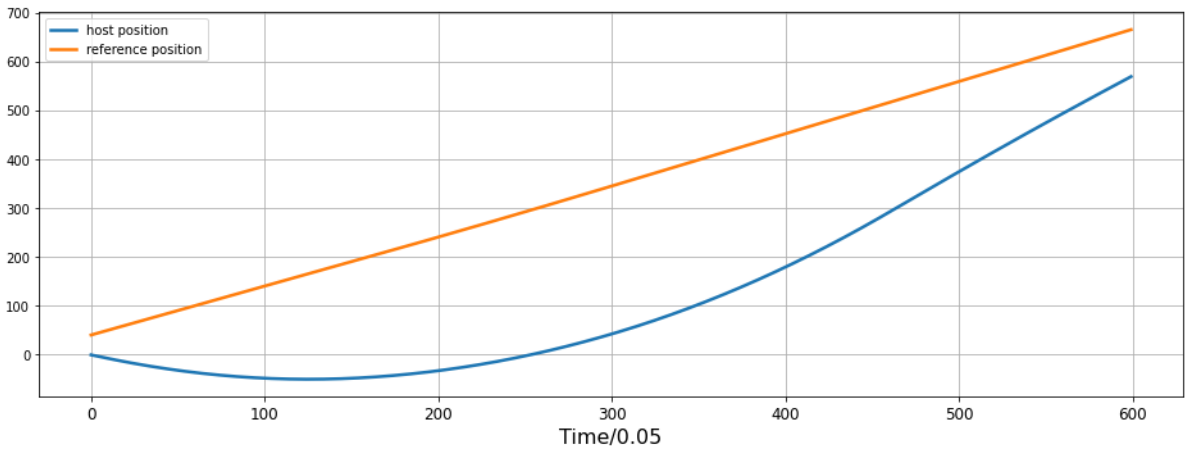


Figure 4.14: In the start host vehicle is at 54 Km/h and the overtaking vehicle is at 72 Km/h.

4.0.3 Lateral motion control - Results

The result of the control space-based approach is shown in figure ???. The obstacles are marked as stars and the track layout is shown separately. The following video further illustrates it. From the video, it is more evident how the car is able to avoid obstacles and track the path effectively.

The lateral motion control is validated by simulation of the algorithm on a rectangular track with static and dynamic obstacles. The track layout is given in 4.15

The trajectory generation is initially evaluated over a single time horizon, incorporating unique features such as varying time steps for computing each sub-trajectory and

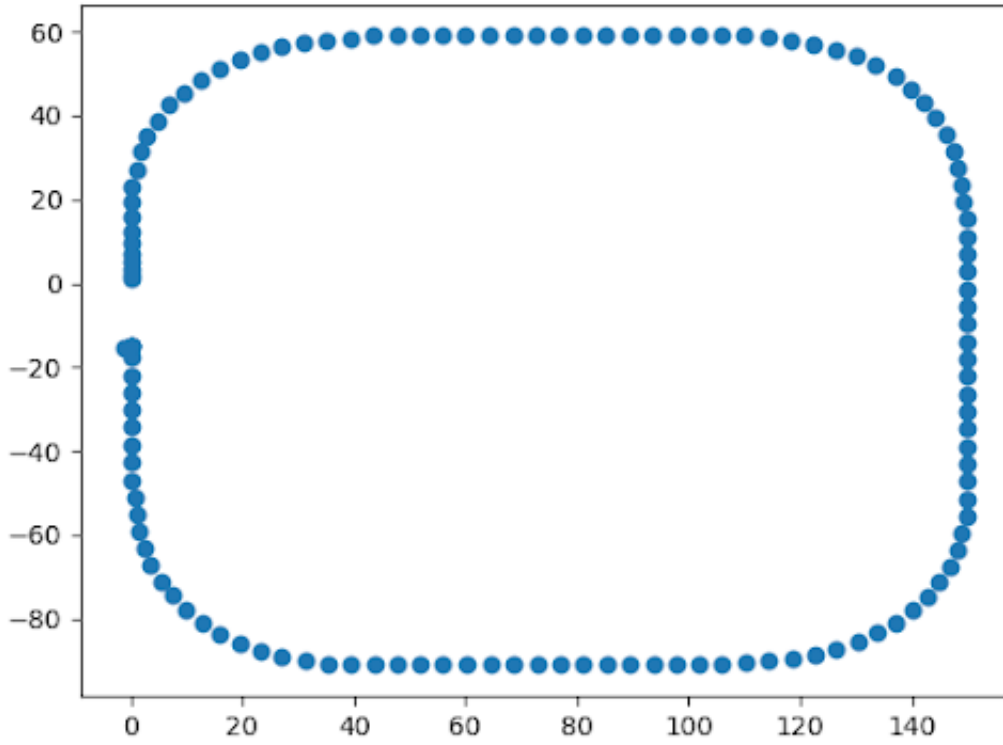
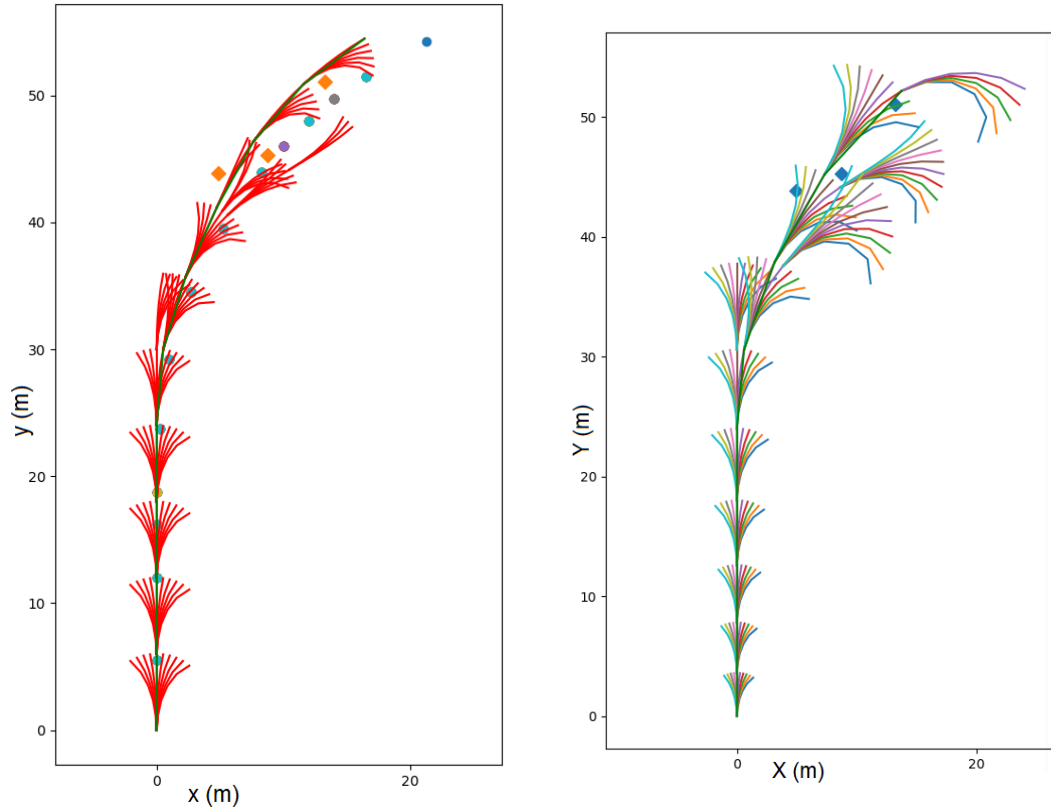


Figure 4.15: The track coordinates corresponding to the centre line.

weights on the costs. These enhancements have significantly improved the results. Figure 4.16 illustrates the effectiveness of the varying time step-based prediction. It shows that the accuracy is higher in the beginning and gradually diminishes towards the end of the prediction horizon. This approach enables longer-term predictions, facilitating the development of smoother trajectories and better collision avoidance strategies.

The inclusion of weights in the cost function to prioritize different action costs has also yielded significant differences. Figure 4.17 showcases the variation in trajectories generated by assigning higher costs to lateral deviation error, steering rate, and steering angle error. The latter trajectory demonstrates a larger margin of avoidance from obstacles as it initiates steering angle changes earlier, thereby reducing the steering rate error. This profile is better suited for real-life driving scenarios where proactive obstacle avoidance is crucial.

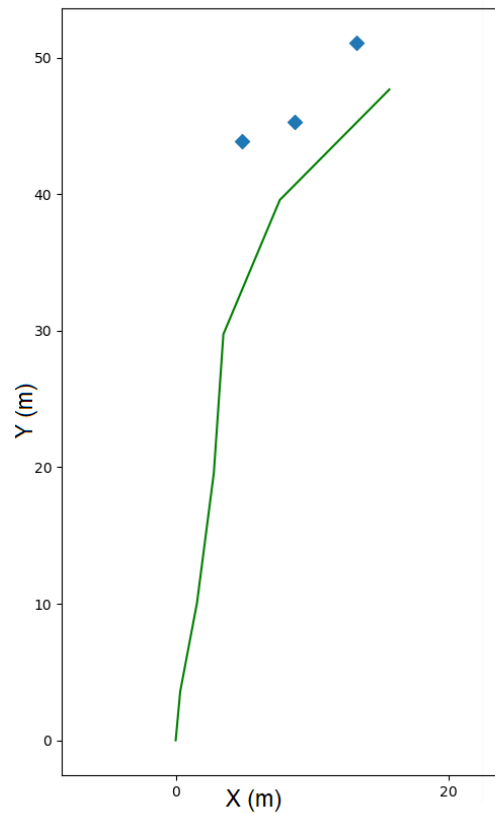
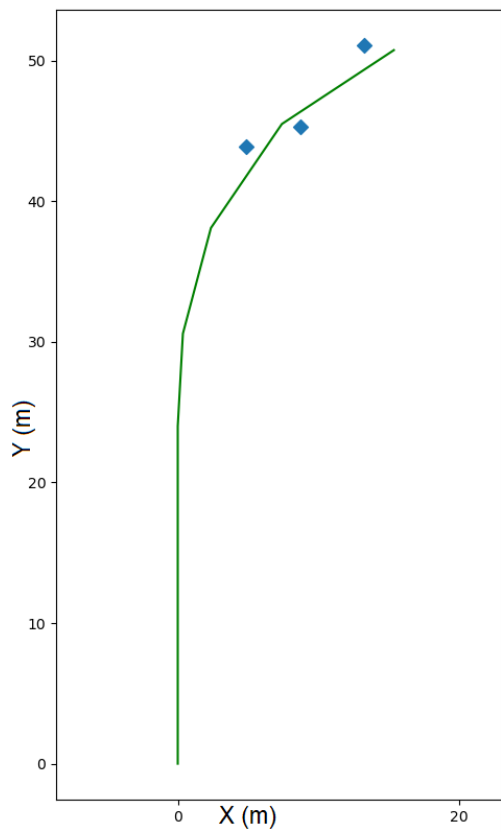
From the result, it is seen that there is an oscillation in the output result. The reasoning for the result is as mentioned. As the obstacles cause the car to take a big turn causing it to move away from the centre line and causing the high cross-track error, it tries to come close to the centre line and since there is a cost associated with the steering input provided the recovery is slow. This causes a low-frequency oscillation upon



(a) state generation without varying time steps and cost. (b) state generation with varying time steps and weights

Figure 4.16: Comparison of trajectory variations - resulting profiles due to variation in varying time steps of planning.

obstacles found on different roadsides. Full results for various weights are shown in this folder. .



(a) High weights on cross-track error resulting in close-to-centre-line trajectory (b) High weights on steering error resulting in a higher deviation from the centre line

Figure 4.17: Comparison of trajectory variations - resulting profiles vary significantly with weights. Also, higher weights on steering errors result in safer driving scenarios

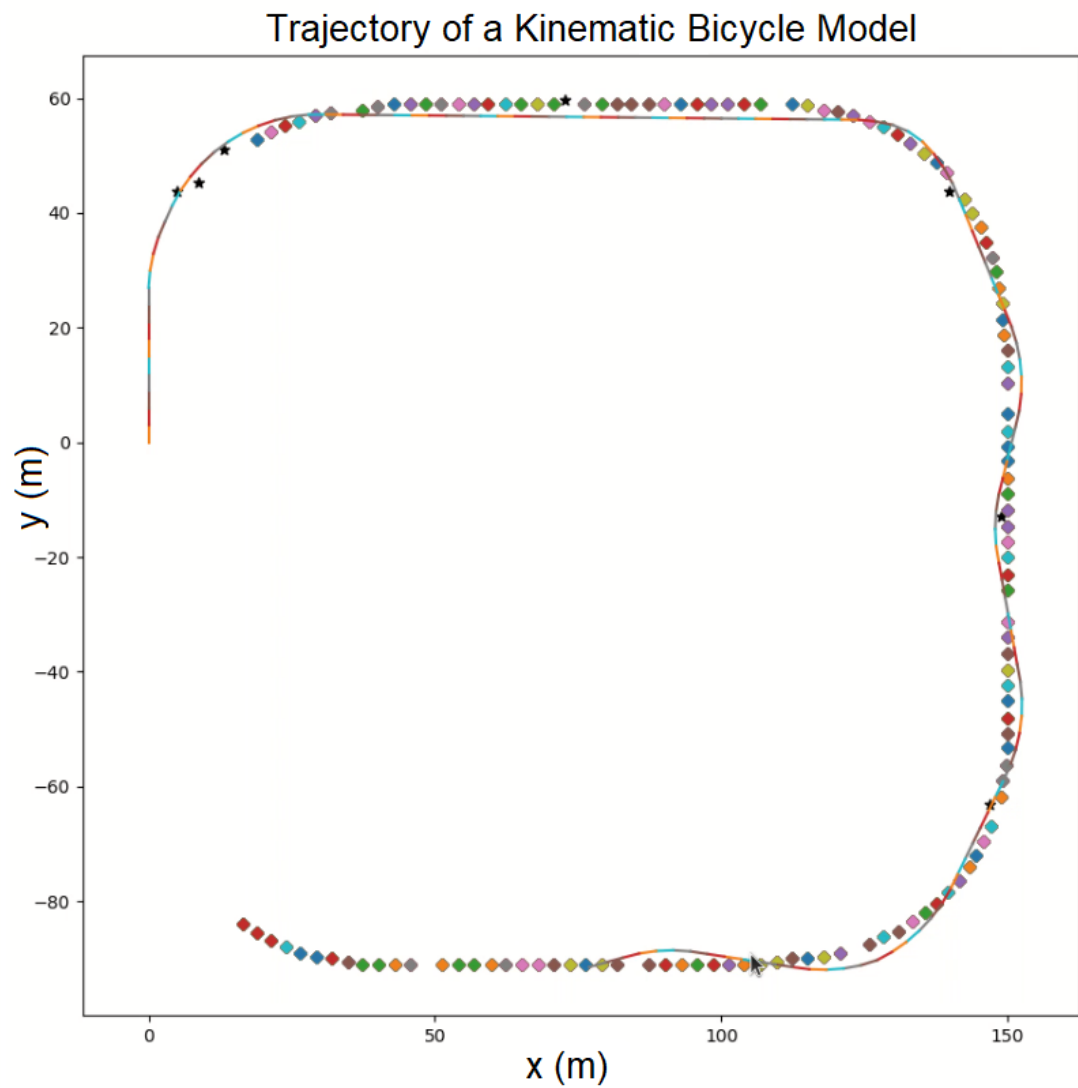


Figure 4.18: Trajectory generation for 420 meters of track. It can be noticed that the generated trajectory is oscillating, this is due to the high cost of steering input

CHAPTER 5

Conclusion and further developemnt

The research aimed to develop and evaluate lateral and longitudinal motion planning algorithms for various scenarios, with the ultimate goal of creating a unified motion planner. The results demonstrate that the lateral motion planning and control components have been successfully achieved in simulation. Computer simulation showed that the MPC is superior to conventional controllers because it treats actuator constraints and has a higher response, the longitudinal motion planning has also shown promising results in specific scenarios such as traffic jam situations and highway overtaking, operating independently. However, further refinements and evaluations are necessary to ensure its effectiveness across a wider range of scenarios and achieve comprehensive motion planning capabilities.

Further work The model predictive controller (MPC) developed for longitudinal control, specifically Adaptive Cruise Control (ACC), will undergo further refinement. The focus will be on enhancing the controller's adaptability to different scenarios. The approach will be to feed the controller with adaptive reference which will also increase the comfort of the driver. The aim is to achieve a controller that can effectively respond to various driving conditions without the need for additional tuning.

The algorithm will undergo testing using a Hardware-in-the-Loop (HIL) setup, which involves integrating it with a dynamometer and utilizing the Webots simulator. This testing approach allows for further development and refinement of the algorithm while adapting it to the specific functionalities and constraints of the hardware platform. By simulating real-world scenarios and incorporating hardware interactions, this HIL testing will provide valuable insights and enable optimization of the algorithm for eventual deployment in physical systems.

use (Taku TAKAHAMA, 2019) or Angel P. de Madrid (2014) or Taku TAKAHAMA (2019) for citing references

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